Homework 1 — Solutions

CE 512 Finite Element Method

2026 Spring

Muhammet Yağcıoğlu

Student No: 290204042

February 26, 2026

İzmir Institute of Technology
Department of Civil Engineering

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1 Introduction

This report solves three 1D element problems: axial truss, cold conduction, and resistive wire. They share the same element geometry and shape functions. Only the field variable, material law, and sign convention differ. Indices $\alpha, \beta \in \{i, j\}$ label the nodes; no summation is implied.

2 Element Geometry, Shape Functions, and General Definitions

2.1 Element domain, boundary, and interpolation

Let (e) denote the oriented 1–simplex with the material coordinate $x \in [0, L]$, node i at $x = 0$, node j at $x = L$, and outward unit normals $n_i = -1$ at $x = 0$, $n_j = +1$ at $x = L$.

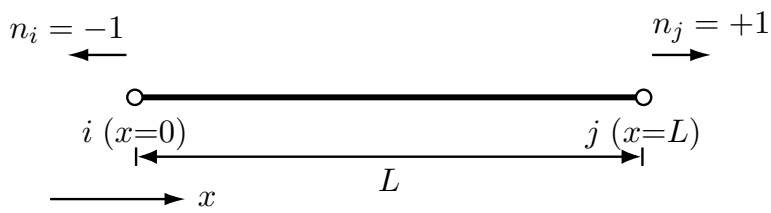

Figure 1: Element (e): domain $[0, L]$ with outward unit normals.

Let $P_1(e)$ be the space of affine functions on e spanned by the shape functions

$$N_i(x) = 1 - \frac{x}{L}, \qquad N_j(x) = \frac{x}{L},$$

satisfying $N_\alpha(x_\beta) = \delta_{\alpha\beta}$ with $x_i := 0$, $x_j := L$. Any field $\phi_h \in P_1(e)$ is $\phi_h(x) = \sum_\alpha N_\alpha(x)\,\phi_\alpha$, $\phi_\alpha := \phi_h(x_\alpha)$.

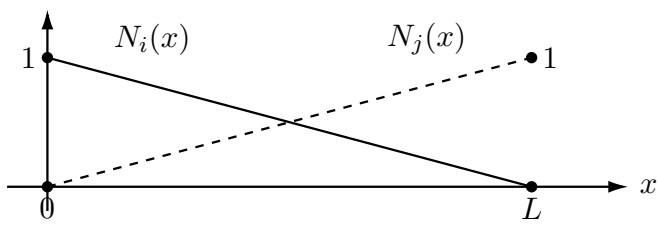

Figure 2: Linear shape functions N_i and N_j on element (e).

2.2 Infinitesimal strain measure and constitutive laws

The one-dimensional infinitesimal (engineering) strain is defined as $\varepsilon(x) := du^{\mathrm{ax}}/dx$, where $u^{\mathrm{ax}}(x)$ is the axial displacement along the positive x-axis. This is the linearized strain measure, valid for $|\varepsilon| \ll 1$, under which the strain–displacement relation is linear, the constitutive law $\sigma = E\varepsilon$ is appropriate, the stiffness matrix is constant, and equilibrium may be written on the undeformed configuration. The constitutive laws for the three problems are:

Problem	**Gradient**	**Constitutive law**	**Material const.**
Truss (axial)	$\varepsilon = \dfrac{du^{\mathrm{ax}}}{dx}$	$\sigma = E\,\varepsilon$ (Hooke)	$E > 0$
Cold conduction	$g = \dfrac{dT}{dx}$	$q^c = k\,g$ (inv. Fourier)	$k > 0$
Electrical	$\mathcal{E} = -\dfrac{dV}{dx}$	$J = \sigma_e\,\mathcal{E}$ (Ohm)	$\sigma_e > 0$

Table 1: Constitutive relations for the three 1-D problems.

3 Problem 1: Truss Member Stiffness Matrix

Let $u^{\mathrm{ax}}(x) \in P_1(e)$ be the axial displacement field. Let the nodal degrees of freedom be the *outward* components of the boundary displacement,

$$u_\alpha := u^{\mathrm{ax}}(x_\alpha)\,n_\alpha, \qquad \alpha \in \{i, j\},$$

so that $u_i = -u^{\mathrm{ax}}(0)$ and $u_j = +u^{\mathrm{ax}}(L)$. Let $\mathbf{u} := \binom{u_i}{u_j}$.

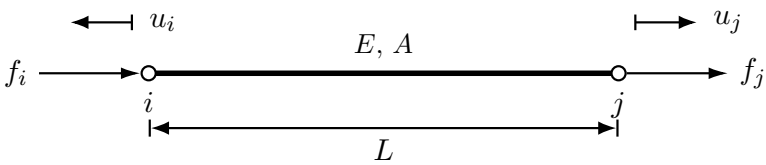

Figure 3: Truss element: DOFs positive outward.

The affine reconstruction consistent with $u^{\mathrm{ax}}(0) = -u_i$ and $u^{\mathrm{ax}}(L) = +u_j$ reads $u_h^{\mathrm{ax}}(x) = -N_i(x)\,u_i + N_j(x)\,u_j$. The infinitesimal axial strain follows as

$$\varepsilon(x) = \frac{du_h^{\mathrm{ax}}}{dx} = \frac{d}{dx}\Big[-\big(1 - \tfrac{x}{L}\big)\,u_i + \tfrac{x}{L}\,u_j\Big] = \frac{u_i + u_j}{L},$$

constant on (e). Let the constitutive law be $\sigma = E\,\varepsilon$ (Hooke) and the axial force $\mathcal{N} = A\,\sigma$, with $E > 0$, $A > 0$ constant. Then $\sigma = \frac{E}{L}(u_i + u_j)$ and $\mathcal{N} = \frac{EA}{L}(u_i + u_j)$.

Let the stored elastic energy be $\mathcal{W}(\mathbf{u}) = \frac{1}{2}\int_0^L EA\,\varepsilon^2\,dx = \frac{1}{2}\,\frac{EA}{L}\,(u_i + u_j)^2$. Let (f_i, f_j) be the energetic conjugates of (u_i, u_j), i.e. $f_\alpha = \partial\mathcal{W}/\partial u_\alpha$. Then

$$f_i = \frac{EA}{L}(u_i + u_j), \qquad f_j = \frac{EA}{L}(u_i + u_j).$$

Let $\mathbf{f} := \binom{f_i}{f_j}$. The element law $\mathbf{f} = K^{(e)}\,\mathbf{u}$ is therefore

$$\boxed{\begin{pmatrix} f_i \\ f_j \end{pmatrix} = \frac{EA}{L}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} u_i \\ u_j \end{pmatrix}.}$$

Column β of $K^{(e)}$ is the force vector when $u_\beta = 1$, $u_\gamma = 0$ $(\gamma \neq \beta)$:

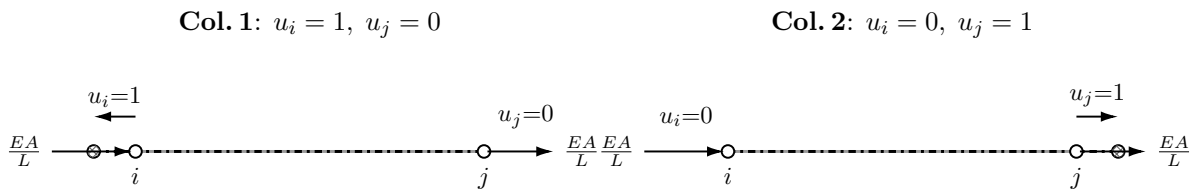

Figure 4: Superposition: unit displacement states → columns of $K^{(e)}$.

$K^{(e)}$ is symmetric ($K^{(e)}_{\alpha\beta} = K^{(e)}_{\beta\alpha}$, Maxwell–Betti reciprocity) and positive semi-definite since $\mathbf{u}^T K^{(e)}\mathbf{u} = \frac{EA}{L}(u_i + u_j)^2 \geq 0$ for all $\mathbf{u}$. It has $\operatorname{rank}(K^{(e)}) = 1$ and $\dim\ker(K^{(e)}) = 1$. A rigid translation $u^{\text{ax}}(x) \equiv c$ gives $u_i = -c$, $u_j = +c$, hence $\mathbf{u}_{\text{rigid}} \propto \binom{-1}{+1}$; the nullspace verification is in Section 7.

4 Problem 2: "Cold" Conduction Matrix

Let $T(x) \in P_1(e)$ be the temperature field with nodal values $T_i := T(0)$, $T_j := T(L)$, and let $\mathbf{T} := \binom{T_i}{T_j}$. The affine approximation and its gradient are

$$T_h(x) = N_i(x)\,T_i + N_j(x)\,T_j, \qquad \frac{dT_h}{dx} = \frac{T_j - T_i}{L}.$$

Let the "cold flux per area" be defined by $q^c(x) = k\,dT_h/dx$, $k > 0$; this is the sign-reversed Fourier law ($q^c = -q^h$), encoding "cold flows from low to high temperature."

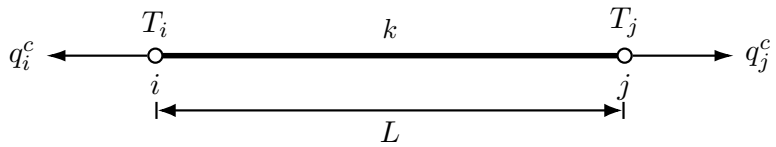

Figure 5: Cold conduction element: $q^c_\alpha > 0$ outward.

Let the outward nodal cold fluxes per area be $q^c_\alpha := q^c(x_\alpha)\,n_\alpha$, so $q^c_\alpha > 0$ means directed out of the element. Since $q^c = k(T_j - T_i)/L$ is constant on (e),

$$q^c_i = q^c\cdot(-1) = \frac{k}{L}(T_i - T_j), \qquad q^c_j = q^c\cdot(+1) = \frac{k}{L}(T_j - T_i).$$

Let $\mathbf{q}^c := \binom{q^c_i}{q^c_j}$. The element cold conduction law $\mathbf{q}^c = K^{(c)}\mathbf{T}$ is

$$\boxed{\begin{pmatrix} q^c_i \\ q^c_j \end{pmatrix} = \frac{k}{L}\begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} T_i \\ T_j \end{pmatrix}.}$$

Let A denote the cross-sectional area; the total cold rates are $\mathbf{Q}^c = A\,\mathbf{q}^c = \frac{kA}{L}\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)\mathbf{T}$. The columns of $K^{(c)}$ arise from unit temperature states:

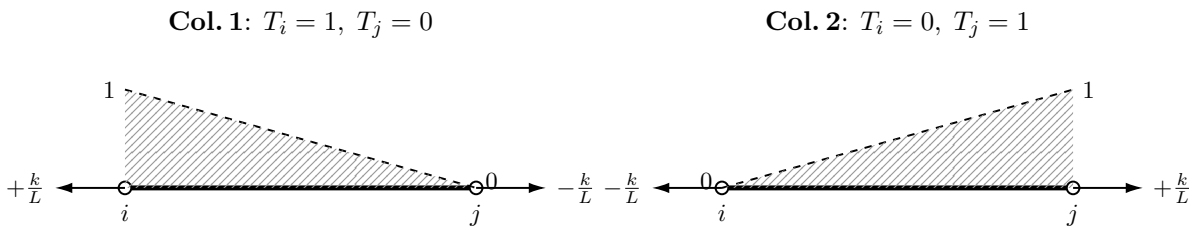

Figure 6: Superposition: unit temperature states → columns of $K^{(c)}$.

$K^{(c)}$ is symmetric and positive semi-definite: $\mathbf{T}^T K^{(c)} \mathbf{T} = \frac{k}{L}(T_i - T_j)^2 \geq 0$. It has $\operatorname{rank}(K^{(c)}) = 1$. A uniform temperature $T_i = T_j$ gives $dT_h/dx = 0$, $\mathbf{q}^c = \mathbf{0}$, so $\binom{1}{1} \in \ker K^{(c)}$; the explicit check is in Section 7.

5 Problem 3: Resistive Wire Current Conduction Matrix

Let $V(x) \in P_1(e)$ be the electric potential with nodal values $V_i := V(0)$, $V_j := V(L)$, and let $\mathbf{V} := \binom{V_i}{V_j}$. Then $V_h(x) = N_i(x)\, V_i + N_j(x)\, V_j$ and $dV_h/dx = (V_j - V_i)/L$. Let the electric field be $\mathcal{E}(x) = -dV_h/dx$ and the current density obey Ohm's law $J(x) = \sigma_e\, \mathcal{E}(x)$, $\sigma_e > 0$, so that

$$J(x) = -\sigma_e \frac{V_j - V_i}{L} = \frac{\sigma_e}{L}(V_i - V_j),$$

constant on (e).

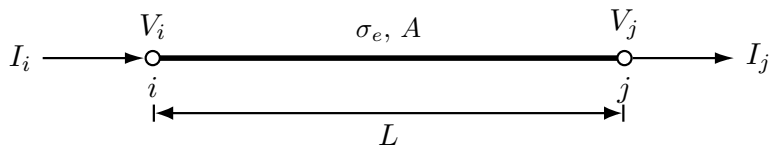

Figure 7: Resistive wire element: I_i into node i, I_j out of node j.

Let the nodal currents be the outward normal fluxes of J over area A: $I_\alpha := A\, J(x_\alpha)\, n_\alpha$. Using $n_i = -1$, $n_j = +1$,

$$I_i = A\, J \cdot (-1) = \frac{\sigma_e A}{L}(-V_i + V_j), \tag{1}$$

$$I_j = A\, J \cdot (+1) = \frac{\sigma_e A}{L}(V_i - V_j). \tag{2}$$

Let $\mathbf{I} := \binom{I_i}{I_j}$. The element current–potential relation $\mathbf{I} = K^{(R)}\mathbf{V}$ is

$$\boxed{\begin{pmatrix} I_i \\ I_j \end{pmatrix} = \frac{\sigma_e A}{L} \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} V_i \\ V_j \end{pmatrix}.}$$

Let $R := L/(\sigma_e A)$ be the resistance and $G := R^{-1} = \sigma_e A/L$ the conductance; the same relation is $\mathbf{I} = G\left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)\mathbf{V}$. The columns of $K^{(R)}$ arise from unit potential states:

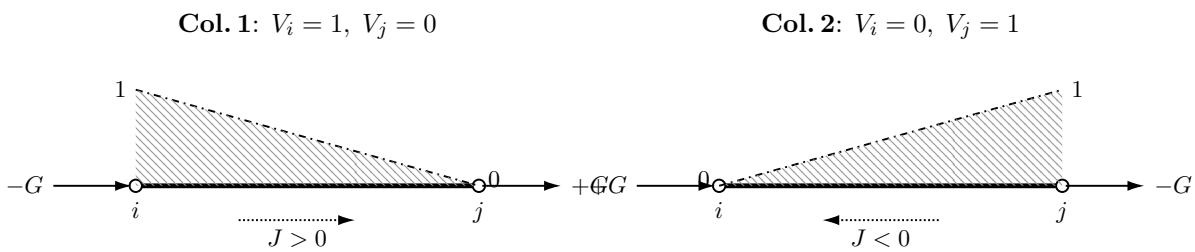

Figure 8: Superposition: unit potential states $\to$ columns of $K^{(R)}$.

$K^{(R)}$ is symmetric and *negative* semi-definite: $\mathbf{V}^T K^{(R)} \mathbf{V} = -G(V_i - V_j)^2 \leq 0$; the negative sign arises because the homework's current convention (into i, out of j) is not a purely outward convention. Under the standard convention $\mathbf{I}^{\text{in}} := -\mathbf{I}$ one recovers the positive semi-definite conductance matrix $\mathbf{I}^{\text{in}} = G\left(\begin{smallmatrix} 1 & -1 \\ -1 & 1 \end{smallmatrix}\right)\mathbf{V}$. The matrix has $\operatorname{rank}(K^{(R)}) = 1$, and a uniform potential $V_i = V_j$ gives $\mathcal{E} = 0$, $J = 0$, confirming $\binom{1}{1} \in \ker K^{(R)}$; see Section 7.

6 Summary

Quantity	**Pr. 1 (Truss)**	**Pr. 2 (Cold Cond.)**	**Pr. 3 (Electrical)**
Primary variable	u (displacement)	T (temperature)	V (potential)
Gradient	$\varepsilon = \dfrac{du}{dx}$	$g = \dfrac{dT}{dx}$	$\mathcal{E} = -\dfrac{dV}{dx}$
Constitutive law	$\sigma = E\varepsilon$	$q^c = kg$	$J = \sigma_e \mathcal{E}$
Material property	E (Young's mod.)	k (therm. cond.)	σ_e (elec. cond.)
Flux / Force	f (force)	q^c (cold flux)	I (current)
Element matrix	$\dfrac{EA}{L}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}$	$\dfrac{k}{L}\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix}$	$\dfrac{\sigma_e A}{L}\begin{pmatrix}-1 & 1\\ 1 & -1\end{pmatrix}$
Nullspace vector	$\begin{pmatrix}-1\\ +1\end{pmatrix}$	$\begin{pmatrix}1\\ 1\end{pmatrix}$	$\begin{pmatrix}1\\ 1\end{pmatrix}$
Physical meaning	rigid translation	uniform temp.	uniform potential

Table 2: Cross-domain analogy for the three 1D finite element problems.

Key observation: The three matrices differ only in sign pattern, a direct consequence of the *different sign conventions* for the degrees of freedom at boundary nodes:

Problem 1: Outward displacement DOF $\Rightarrow$ sign flip at node i $\Rightarrow$ all-positive matrix. **Problem 2:** Standard nodal values, outward flux $\Rightarrow$ classical symmetric positive semidefinite conduction matrix. **Problem 3:** Into-i/out-of-j current convention $\Rightarrow$ negative semidefinite; flip to "into-element" convention recovers the standard form.

7 Validation: Nullspace (Zero-Gradient Mode) Checks

Each element matrix must annihilate the DOF vector corresponding to the trivial (zero-gradient) state of the respective field. The three verifications are as follows.

Problem 1 — Rigid-body translation. A rigid translation $u^{\text{ax}}(x) \equiv c$ gives $u_i = n_i\, c = -c$ and $u_j = n_j\, c = +c$, hence $\mathbf{u}_{\text{rigid}} \propto \binom{-1}{+1}$ and $\varepsilon = (u_i + u_j)/L = 0$. Substituting into the element law:

$$K^{(e)}\begin{pmatrix}-1\\ +1\end{pmatrix} = \frac{EA}{L}\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}\begin{pmatrix}-1\\ +1\end{pmatrix} = \frac{EA}{L}\begin{pmatrix}-1+1\\ -1+1\end{pmatrix} = \begin{pmatrix}0\\ 0\end{pmatrix}. \quad \checkmark$$

Problem 2 — Uniform temperature. A uniform temperature $T_i = T_j$ gives $dT_h/dx = (T_j - T_i)/L = 0$, hence $q^c = 0$ and $\mathbf{q}^c = \mathbf{0}$. The nullspace vector is $\binom{1}{1}$:

$$K^{(c)}\begin{pmatrix}1\\ 1\end{pmatrix} = \frac{k}{L}\begin{pmatrix}1 & -1\\ -1 & 1\end{pmatrix}\begin{pmatrix}1\\ 1\end{pmatrix} = \frac{k}{L}\begin{pmatrix}1-1\\ -1+1\end{pmatrix} = \begin{pmatrix}0\\ 0\end{pmatrix}. \quad \checkmark$$

Problem 3 — Uniform potential. A uniform potential $V_i = V_j$ gives $dV_h/dx = 0$, $\mathcal{E} = 0$,

$J = 0$, and no current flows. The nullspace vector is $\binom{1}{1}$:

$$K^{(R)}\begin{pmatrix}1\\1\end{pmatrix} = \frac{\sigma_e A}{L}\begin{pmatrix}-1 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1\\1\end{pmatrix} = \frac{\sigma_e A}{L}\begin{pmatrix}-1+1\\ 1-1\end{pmatrix} = \begin{pmatrix}0\\0\end{pmatrix}. \quad \checkmark$$

All three matrices are confirmed to be rank-1, symmetric, and to possess a physically meaningful one-dimensional nullspace corresponding to the zero-gradient mode of the respective field.

8 Discussion: Sign Convention and Its Implication on the Kernel

The choice of positive direction for the nodal degrees of freedom is not merely a labelling convention; it directly determines the sign pattern of the element stiffness matrix and, consequently, the algebraic form of its kernel vector. This section discusses this relationship for all three problems.

In Problem 1 the DOFs are defined as the *outward* displacement components $u_\alpha = u^{\text{ax}}(x_\alpha)\, n_\alpha$. Because $n_i = -1$, the DOF at node i has the opposite sign of the physical displacement. A rigid translation $u^{\text{ax}} \equiv c$ therefore maps to $u_i = -c$, $u_j = +c$, giving the kernel vector $\binom{-1}{+1}$. This sign flip at node i is also the reason why $K^{(e)}$ has all-positive entries: both columns produce force in the same direction.

In Problem 2 the DOFs are simply the nodal temperatures T_i, T_j, with no sign reversal at either node. A uniform temperature $T_i = T_j = c$ gives the kernel vector $\binom{1}{1}$. The matrix $K^{(c)}$ has the classical pattern $\left(\begin{smallmatrix}+1 & -1\\ -1 & +1\end{smallmatrix}\right)$, positive semi-definite, because the DOF convention and the flux convention are both standard outward.

In Problem 3 the DOFs are again plain nodal values V_i, V_j, so a uniform potential gives $\binom{1}{1} \in \ker K^{(R)}$, just like Problem 2. However, the homework specifies current *into* node i and *out of* node j, which is not a purely outward convention. This mixed sign in the current direction introduces a global sign flip: $K^{(R)} = -G\left(\begin{smallmatrix}1 & -1\\ -1 & 1\end{smallmatrix}\right)$, making it negative semi-definite. If one redefines $\mathbf{I}^{\text{in}} := -\mathbf{I}$ (all currents into the element), the standard positive semi-definite form is recovered.

Table 3 collects the sign convention, the resulting matrix definiteness, and the kernel vector for each problem.

	Pr. 1 (Truss)	**Pr. 2 (Cold Cond.)**	**Pr. 3 (Electrical)**
DOF definition	$u_\alpha = u^{\text{ax}} n_\alpha$	$T_\alpha = T(x_\alpha)$	$V_\alpha = V(x_\alpha)$
Sign convention	outward disp.	nodal value	nodal value
Flux / force conv.	outward	outward	into-i, out-of-j
Figure	Figure 3	Figure 5	Figure 7
Element matrix sign	all +	$\left(\begin{smallmatrix}+ & -\\ - & +\end{smallmatrix}\right)$	$\left(\begin{smallmatrix}- & +\\ + & -\end{smallmatrix}\right)$
Definiteness	pos. semi-def.	pos. semi-def.	neg. semi-def.
Kernel vector	$\begin{pmatrix}-1\\+1\end{pmatrix}$	$\begin{pmatrix}1\\1\end{pmatrix}$	$\begin{pmatrix}1\\1\end{pmatrix}$
Zero-gradient mode	rigid translation	uniform temp.	uniform potential

Table 3: Sign convention, matrix definiteness, and kernel vector for each problem.

The key insight is that all three kernels encode the same physics: a spatially constant field produces zero gradient and therefore zero internal force or flux. The algebraic form of the kernel vector, however, depends on how the DOFs relate to the physical field. When the DOF includes a sign reversal (as in Problem 1), the kernel vector inherits that reversal.

9 Conclusion

All three matrices were derived step by step. They are all rank-1 and symmetric. Each has a nullspace with a clear physical meaning. The sign patterns come from the DOF conventions. The results were checked in Section 7.